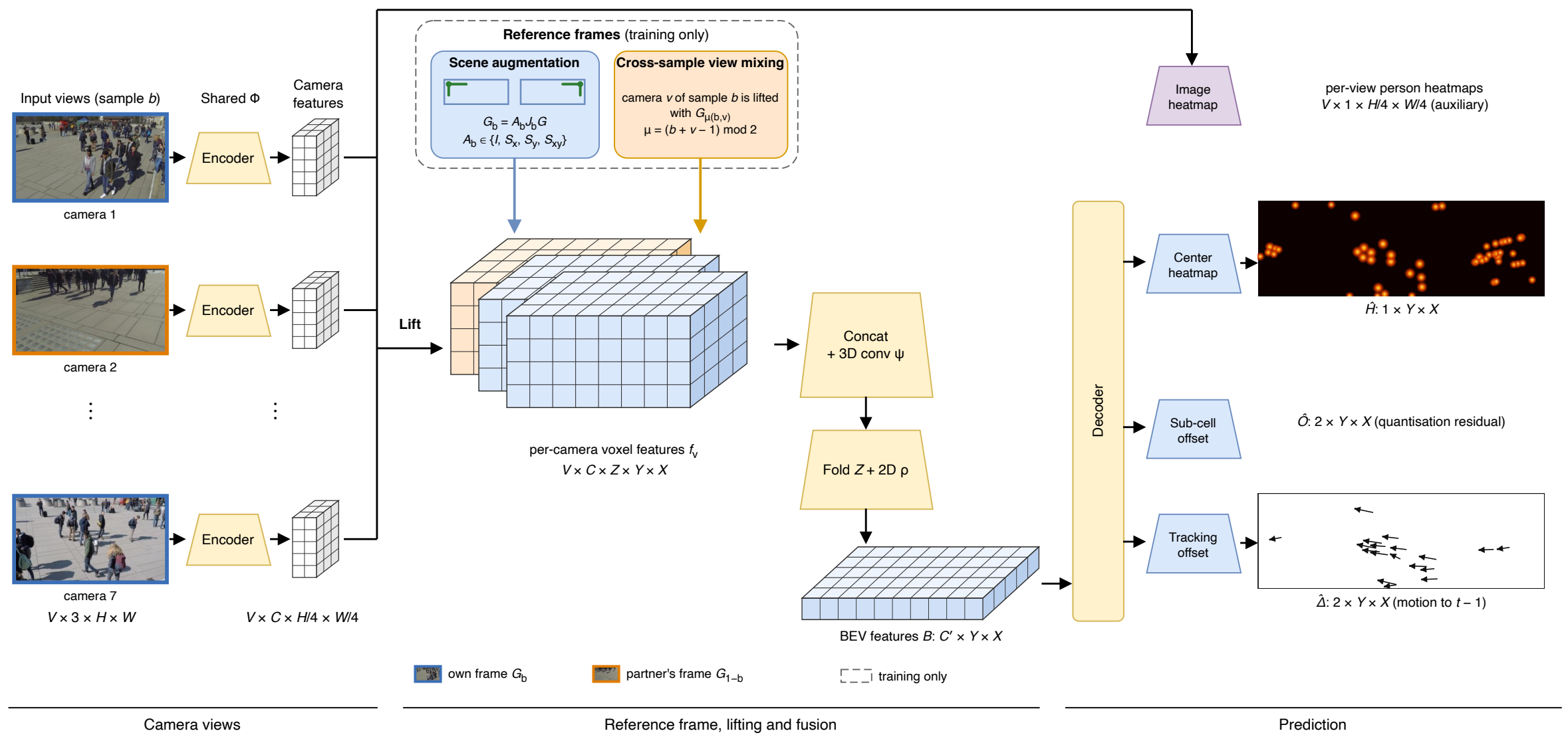

Input views (sample b)
Shared Φ
Camera features
camera 1
camera 2
camera 7
V × 3 × H × W
V × C × H/4 × W/4
Encoder
Encoder
Encoder
Reference frames (training only)
Scene augmentation
G_b = A_b J_b G
A_b ∈ {I, S_x, S_y, S_xy}
Cross-sample view mixing
camera v of sample b is lifted with G_μ(b,v)
μ = (b + v − 1) mod 2
Lift
per-camera voxel features f_v
V × C × Z × Y × X
Concat + 3D conv ψ
Fold Z + 2D ρ
BEV features B: C′ × Y × X
Decoder
Image heatmap
per-view person heatmaps V × 1 × H/4 × W/4 (auxiliary)
Center heatmap
Ĥ: 1 × Y × X
Sub-cell offset
Ô: 2 × Y × X (quantisation residual)
Tracking offset
Δ̂: 2 × Y × X (motion to t − 1)
own frame G_b
partner's frame G_1−b
training only
Camera views
Reference frame, lifting and fusion
Prediction